MIS: Difficulty Scoring as a Pre-Augmentation Diagnostic for Agricultural Object Detection

Claim (reframed from original draft). We propose: MIS a score that can provides structural diagnostics that identify the structural conditions underlying copy-paste augmentation failure before augmentation is conducted. The claim is not that MIS improves detection performance. It changed to MIS exposes *why* copy-paste fails on a given dataset, which is useful before wasting compute on augmentation experiments.

Why the claim changed. The original goal was to show MIS-guided copy-paste improves AP. This belief was based on the research of Kisantal et al. [1], When reanalyzing the result, i believe their success relied on 2 specific conditions: (a) the MS COCO dataset with 3–5 objects per image on average, (b) a two-stage Faster R-CNN with anchor-based RPN. That make my assumstion that the method will work on different agricultural datasets has low chance of success. Because of that, rather than searching for a positive result on a new dataset, I chose to understand *why* the failure is consistent and whether MIS identifies that reason.

Methodology

1.1 MIS Scoring

Predictions generated at $\tau_{\text{conf}} = 0.001$; matched to ground truth at $\tau_{\text{IoU}} = 0.5$. The **object difficulty score**:

$$S_{\text{obj}}(g) = \begin{cases} \min_{p \in \mathcal{P}(g)} \left[\alpha(1 - c_p) + \beta(1 - u_p)\right], & \mathcal{P}(g) \neq \varnothing \\ \alpha + \beta = 1.0, & \text{missed object} \end{cases} \tag{1}$$

$\alpha = \beta = 0.5$, $c_p = \text{conf}(p)$, $u_p = \text{IoU}(p, g)$. Taking the minimum removes noise from the many low-confidence candidates generated at $\tau_{\text{conf}} = 0.001$.

The **image difficulty score**:

$$S_{\text{img}}(I) = w_1 \cdot \frac{1}{|G(I)|} \sum_{g \in G(I)} S_{\text{obj}}(g) + w_2 \cdot \text{FPrate}(I) \tag{2}$$

$w_1 = 1$ (fixed). w_2 auto-calibrated by grid search to minimize $|r(S_{\text{img}}, \text{miss\_rate}) - r(S_{\text{img}}, \text{FP\_rate})|$.

1.2 MinneApple Dataset Split Problem and Fix

Problem I found late. MinneApple contains only 10 unique trees photographed at different times. The test set was captured approximately one year after training. A random 80/20 image-level split places images from the same tree in both training and validation, causing the model to memorize tree-specific features. This inflated my baseline to AP = 0.439 (contaminated; discarded).

Fix. I try to implement the split that authors used on there original paper, Tree-identity-clean split: 30 validation images from trees not in training (≈3 per held-out tree). Training: 200 epochs max, early stopping patience 50. Clean baseline: AP = 0.353 ± 0.005.

1.3 MinneApple Size Thresholds

As i analyse the Minne Apple dataset, I see that the COCO thresholds (small ≤ 32 px, medium ≤ 96 px) do not suit (traning set has no large objects and the test set has some of it which make the AP score following the size category not meaningful). I define thresholds calibrated to the training distribution:

Category	Criterion	Train %
Very Small	area $\leq 400\,\text{px}^2$ ($< 20\times20$ px)	23.08%
Small	$400 <$area$\leq 1024\,\text{px}^2$ (20–32 px)	50.53%
Medium	area$> 1024\,\text{px}^2$ (> 32 px)	26.39%

1.4 Augmentation Conditions

Baed on the advices, I try a new method of copy and paste augmentation that is easier to analyze.

First, all training images are ranked using the image-level difficulty score and divided into three equal groups: Score High, Score Medium, and Score Low. Depending on the experiment, images and objects are sampled from one of these groups.

To introduce some variation, each pasted object is randomly transformed with a scaling factor between 0.9 and 1.1, and a rotation of up to ±15 degrees. The objects are pasted directly without any blending. Training set extended by 30% (189 synthetic images, same-image constraint: objects pasted into their source image to preserve photometric consistency. Objects pasted: 2–8 per image, scale $[0.9, 1.1]$, rotation $\pm 15°$, no blending. [1]). Each image can be selected for augmentation at most three times, and each object can also be reused at most three times across the entire dataset.

Condition	**Image pool**	**Object pool**
Random	Uniform random	Uniform random
Score High	Top tertile	Top tertile
Score Medium	Middle tertile	Middle tertile
Score Low	Bottom tertile	Bottom tertile

Each condition: 3 seeds.

Score Validation

2.1 Image-Level Correlation

Table 1: Pearson correlation of S_{img} with per-image error rates.

Dataset	r(score, miss)	r(score, FP)
MinneApple	0.698	0.688
GWHD 2021	0.816	0.818

Moderate-to-strong on both. GWHD is higher because difficulty is structured at the domain level, giving a cleaner signal.

2.2 Object-Size Validity: MinneApple

Table 2: Mean S_{obj} and baseline AP per size category. Consistent across all 3 seeds. Spearman $r_s = -0.82$: strong monotonic inverse relationship.

Category	**AP**	**Mean S**	**Std**	**In top 30% score tertile**
Very Small	0.071	0.1524	0.1433	**64.39%** (5138 objects)
Small	0.270	0.0755	0.0475	33.88% (2704 objects)
Medium	0.493	0.0526	0.0184	**1.73%** (138 objects)

My observation. 98.27% of high-difficulty objects are in size categories where the model already performs poorly (AP ≤ 0.27). Only 1.73% are in the medium category where the model has room to improve. This means score-guided augmentation, regardless of which difficulty tier it targets, almost never selects objects in the learnable zone. This is the key diagnostic finding on MinneApple.

2.3 Domain-Level Validity: GWHD 2021 (18 domains)

Note on dataset version. I initially used a 7-domain subset of GWHD. I re-ran on GWHD 2021 with 18 domains. The correlation is stronger on the full dataset: Pearson $r = -0.905$, Spearman $r_s = -0.899$.

Table 3: GWHD 2021 domain analysis: MIS median difficulty vs. AP (18 domains, sorted by AP descending). MIS ranks all 18 domains correctly without domain labels.

Domain	**Med S**	**AP**	**Domain**	**Med S**	**AP**
Arvalis_10	0.200	0.744	Arvalis_8	0.342	0.546
Rres_1	0.227	0.658	Arvalis_5	0.342	0.542
ETHZ_1	0.268	0.643	Arvalis_9	0.372	0.520
NMBU_2	0.299	0.639	Arvalis_12	0.395	0.472
Inrae_1	0.264	0.633	Arvalis_3	0.350	0.470
Arvalis_11	0.358	0.606	Arvalis_7	0.407	0.463
NMBU_1	0.335	0.601	Arvalis_2	0.386	0.454
Arvalis_6	0.289	0.600	Arvalis_1	0.400	0.452
ULiège-GxABT_1	0.380	0.597	Arvalis_4	0.475	0.387

3.1 MinneApple: Structural Bottleneck Pattern

Table 4: MinneApple: object scale distribution and 3-seed mean AP. All ΔAP are within one standard deviation of baseline (ns).

Condition	V.Small	Small	Medium	Gap	AP$\pm\sigma$	ΔAP
Test target	17.17%	38.14%	44.70%	0 pp	—	—
Baseline	23.08%	50.53%	26.39%	−18.31 pp	0.353±0.005	—
Random CP	23.52%	49.98%	26.50%	−18.20 pp	0.346±0.005	−0.007
Score High	26.48%	49.59%	23.92%	−20.78 pp	0.350±0.007	−0.003
Score Medium	22.56%	50.80%	26.64%	−18.06 pp	0.348±0.006	−0.005
Score Low	19.99%	48.56%	31.45%	−13.25 pp	0.348±0.003	−0.005

My observation on the Score Low result. Score Low achieves the largest distribution shift of any condition (medium gap narrows from 18.31 to 13.25 pp) yet produces no improvement. This is the most informative result because it eliminates selection strategy as the explanation. Even when I select the easiest, most medium-sized objects, the augmentation density increase is only 3.4% (42.57 → 44.03 objects per image). On a dataset already averaging 42 objects per image, a 3.4% density increase is insufficient to shift model behavior. The failure is a **structural bottleneck**, not a selection problem: the dataset is too dense for copy-paste to add meaningful new signal.

This is in direct contrast to Kisantal et al. [1], who achieved 30–60% density increases starting from 3–5 objects per image. The mechanism that produced their gains cannot operate at MinneApple's baseline density.

Feature space confirmation. From image 1.

YOLOv8s P5 layer PCA centroids: original train PC1= +4.83, augmented PC1= +3.72, test PC1= −11.24. Augmented images remain co-located with original training data (ΔPC1 ≈ 1.1), both separated from test by ΔPC1 ≈ 16. Copy-paste does not move the feature distribution toward the test set.

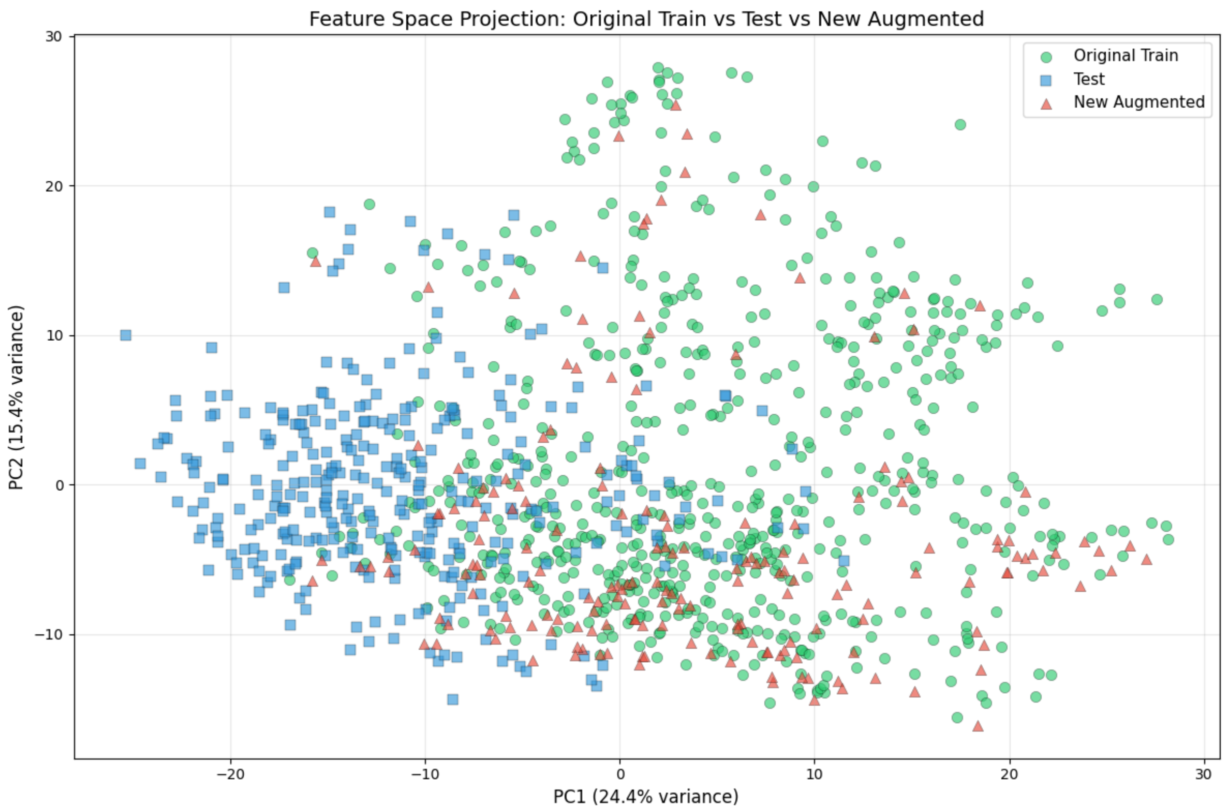

Figure 1: Feature space distribution of original training, augmented training, and test images in MinneApple.

3.2 GWHD 2021: Stochastic Instability Pattern

The GWHD results are qualitatively different from MinneApple and reveal a second, distinct failure mode.

Table 5: GWHD 2021 Score High augmentation results per seed (mean AP, Score High condition only).

Seed	**Baseline AP**	**Augmented AP**
Seed 456	0.2126	0.2549 **+4.2 AP vs. baseline (helped)**
Seed 123	0.2307	0.1755 **−5.5 AP vs. baseline (hurt badly)**
Seed 789	0.2296	0.2239 (−0.6 AP, no effect)
Mean ± std	0.2296	0.2239 ± 3.98%

My observation on GWHD seed variance. The ±3.98% standard deviation across seeds is large relative to the mean effect (−0.006 AP). This is not random noise — it is structurally caused. MIS identifies that high-difficulty images in GWHD concentrate in minority domains (Arvalis group, lowest AP). When augmentation targets those images, it introduces training signals dominated by a small number of underrepresented domains. Whether that helps or hurts depends on which random seed determines the sampling order, resulting in high-variance outcomes.

Why this is different from MinneApple. On MinneApple, all five conditions converge to near-zero ΔAP with low seed variance ($\sigma \leq 0.007$). The bottleneck is structural: density and resolution limits act consistently regardless of seed. On GWHD, the bottleneck is distributional: minority-domain concentration creates inconsistent training signals that vary with sampling order.

Important implication. For a practitioner: GWHD-type failure is actually *worse* than MinneApple-type failure. A dataset with structural bottlenecks will consistently not improve; you can measure this and move on. A dataset with stochastic instability might improve with one seed and fail badly with another — the augmentation is unreliable rather than simply ineffective.

Diagnostic Framework

What this framework is and is not. The framework is a structured way to use MIS scores to reason about whether copy-paste augmentation is likely to help, before running any augmentation experiments. It is **not a formal prediction system** with guaranteed accuracy. The criteria were observed from two datasets (MinneApple and GWHD 2021) and are retrospectively validated on the same data. A prospective test on an unseen dataset would be needed to establish it as a predictive tool. I present it as an *analytical checklist* for practitioners.

4.1 The Three Structural States

Rather than hard numerical thresholds (which would be arbitrary given only two datasets), I frame each criterion as a qualitative structural state that MIS can identify.

State 1 — Information Bottleneck.
Definition: High-difficulty instances, as identified by MIS, reside in a category where baseline AP is near zero. This indicates a hardware/resolution limit rather than a representation shortage: the model has no useful gradient to receive from these instances regardless of how many times they appear in training.

How to check with MIS:

1. Compute S_{obj} for all training objects.
2. Find the top-difficulty tertile.
3. Check what size/domain category dominates that tertile.
4. Look up the baseline AP for that category.
5. If AP is near zero — the model fails on these objects almost completely — this state is present.

Action: Copy-paste augmentation concentrated on high-difficulty objects (Score High) will predominantly paste information-bottleneck instances. **Score High augmentation is unlikely to help.**

MinneApple result: 64.39% of hard objects are very small (AP$=0.071$). Information bottleneck state confirmed.

State 2 — Stochastic Instability Risk.
Definition: High-difficulty images, as identified by MIS, are concentrated in minority domains. This means score-guided augmentation will disproportionately sample from those domains, creating high-variance training signals whose effect depends on sampling order.

How to check with MIS:

1. Find the top-difficulty tertile by S_{img}.
2. Check which domains those images come from.
3. If they are dominated by a small number of minority domains (domains with low domain-level AP), this state is present.

Action: Score-guided augmentation is not just unlikely to help — it introduces **high risk of stochastic instability**: results may improve for some training seeds and degrade for others. This makes augmentation *unreliable* rather than simply ineffective.

GWHD 2021 result: Hard images concentrate in Arvalis minority domains. Stochastic instability confirmed: seed 456 +4.2 AP, seed 123 −5.5 AP ($\sigma = 3.98\%$).

State 3 — Grid Saturation.
Definition: The dataset's baseline object density is high enough that copy-paste augmentation at a standard ratio (e.g., 30%) produces a negligible proportional increase in object density. When the baseline is already dense, pasting 5 more objects into an image that already contains 42 does not meaningfully shift the training distribution.

How to check (no MIS required):

1. Compute mean objects per image in the training set.
2. Estimate density increase: $\frac{\text{pasted objects per image} \times \text{augmentation ratio}}{\text{mean objects per image}}$.
3. If the increase is small relative to baseline density, the dataset is in grid saturation for copy-paste at this ratio.

Action: Either significantly increase the augmentation ratio, use cross-image paste (which this work did not test), or accept that copy-paste cannot shift the distribution on this dataset at this ratio.

MinneApple result: $(5 \times 0.35)/42.57 \approx 4.1\%$ density increase. Score Low confirmed this: even the condition that moved the distribution most (31.45% medium vs. 26.39% baseline) produced no AP improvement.

4.2 Summary Diagnostic Table

Table 6: Applying the three structural states to MinneApple and GWHD 2021. All three states are present on both datasets. Predictions are consistent with experimental outcomes.

State	MinneApple	GWHD 2021	Observed
Information Bottleneck: hard objects near AP$=0$	64.39% of hard objects very small (AP$=0.07$). State: **present**	AP varies by domain; not primary state.	Structural bottleneck: consistent near-zero ΔAP ($\sigma \leq 0.007$)
Stochastic Instability: hard images in minority domains	Single domain dataset; not applicable.	Hard images in Arvalis minority group (lowest AP). State: **present**	High variance: $\pm 3.98\%$ across seeds
Grid Saturation: negligible density increase	4.1% increase on 42.57 obj/img. State: **present**	Similar pattern.	Distribution shift insufficient
Overall prediction	**Augmentation unlikely to help reliably**	**Augmentation unreliable (high variance)**	**Both confirmed**

4.3 When Copy-Paste Is Likely to Help

The framework implies three conditions that together indicate copy-paste augmentation is worth attempting: (a) hard objects reside in a category where the model has partial but not saturated performance — neither near-zero nor already strong; (b) high-difficulty images are spread across domains rather than concentrated in a minority; (c) augmentation meaningfully increases object density relative to the baseline.

These conditions are consistent with Kisantal et al.'s success on COCO and are absent on both MinneApple and GWHD 2021.

Known Limitations and What I Would Do With More Time

1. **Prospective validation.** The diagnostic framework was derived from the same datasets used to validate it. Testing it on a dataset where copy-paste is expected to help (e.g., a sparse, single-domain dataset) and observing that the framework correctly identifies the absence of all three bottleneck states would substantially strengthen the claim.
2. **Threshold formalization.** The three structural states are described qualitatively. With more datasets spanning the full spectrum from failing to succeeding, the thresholds for each state could be derived

statistically rather than by observation. With two failing datasets, no such derivation is valid.

3. **Using score with more advanced generative augmentation methods.** This work only tests copy-paste, but the diagnostic framework could be applied to other methods (e.g., generative models) that also have a selection component. It would be interesting to see if the same bottleneck states apply or if new ones emerge with different augmentation techniques.

References

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